

Database Schema Documentation

This document provides a detailed overview of the database schema for the Libro Library Management System.

Tables

The project uses three main tables:

1. Libro_books

Stores information about the books available in the library.

Column	Type	Description
id	INT	Primary key, auto-incremented.
title	VARCHAR(255)	Title of the book.
author	VARCHAR(255)	Author of the book.
isbn	VARCHAR(100)	Unique International Standard Book Number.
quantity	INT	Total quantity of copies available.
shelf_location	VARCHAR(100)	Location of the book in the library shelves.

Indexes:

- idx_title (title)
- idx_author (author)
- idx_isbn (isbn) - Unique

2. Libro_borrowers

Stores information about the library members.

Column	Type	Description
id	INT	Primary key, auto-incremented.
name	VARCHAR(255)	Name of the borrower.
email	VARCHAR(255)	Unique email address.
registered_date	DATE	Date the borrower registered.

Indexes:

- idx_name (name)
- idx_email (email) - Unique

3. Libro_BorrowingProcess

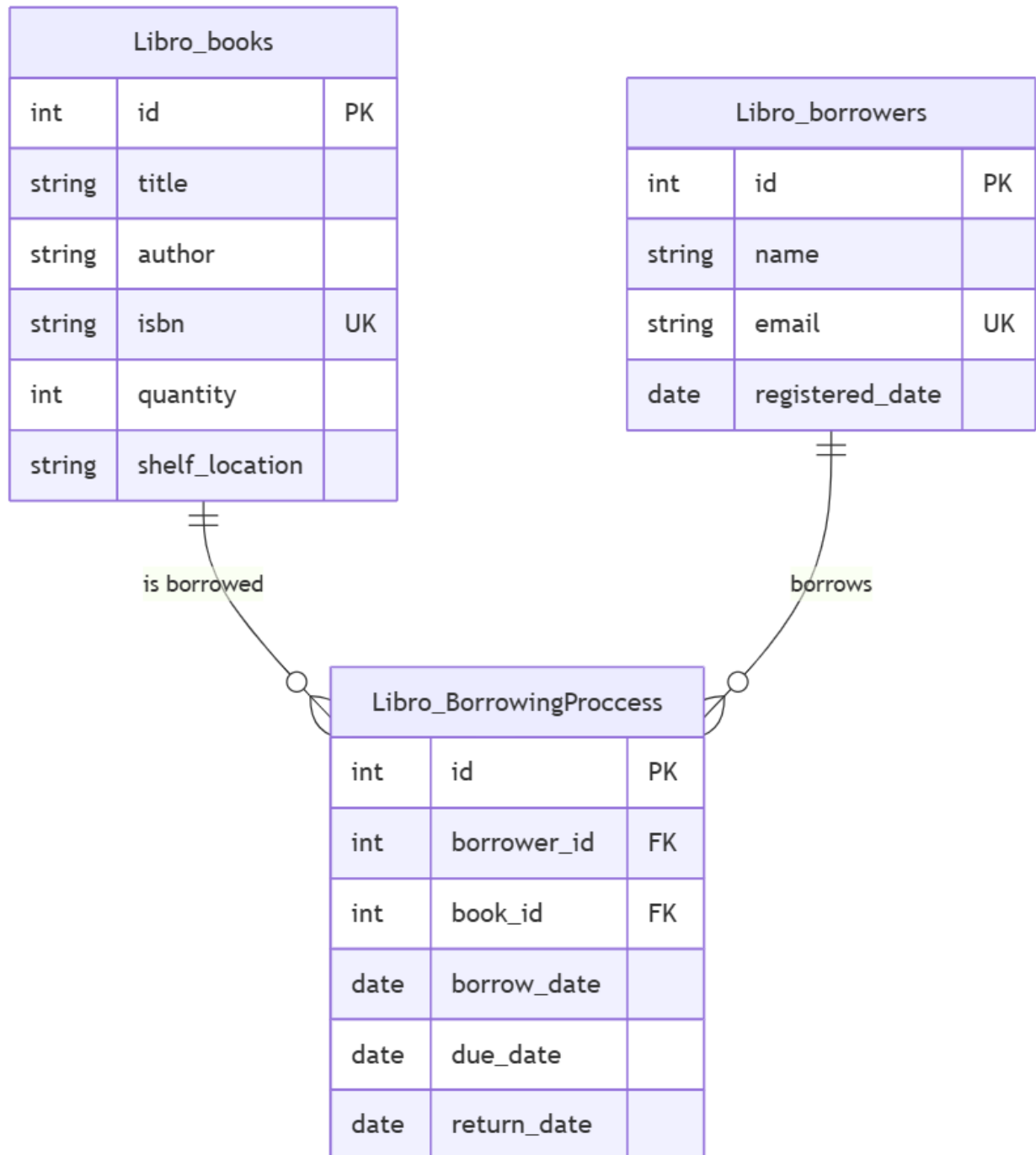
Tracks the borrowing and return of books.

Column	Type	Description
id	INT	Primary key, auto-incremented.
borrower_id	INT	Foreign key referencing Libro_borrowers(id).
book_id	INT	Foreign key referencing Libro_books(id).
borrow_date	DATE	Date the book was borrowed.
due_date	DATE	Expected return date.
return_date	DATE	Actual return date (NULL if not yet returned).

Indexes:

- idx_borrower_id (borrower_id)
- idx_book_id (book_id)
- idx_due_date (due_date)

Schema Diagram (ERD)



Setup Instructions

To set up the database, execute the SQL script in [SCHEMA_QUERY.SQL](#) on your MySQL server.

1. Connect to your MySQL server.

2. Create a new database (e.g., `CREATE DATABASE libro;`).
3. Run the provided script within the `libro` database context.